- ZIGZAG CONVERSION:

i/p: "PAYPALISHIRING", n=3

o/p: "PAHNAPLSIIGYIR"

Approach:

→ if n==1, return string directly

→ create 2D array

→ traverse input string

↳ place in each row

↳ use flag & down or up to decide whether to move down or up rows.

→ After traversal, convert to string and return string.

Pseudocode:

```
if (n==1){
    print(s);
    return;
}
String[] rows
currRow=0
down= true
for (0 to s.length()) → i
{   ch = charAt(i)
    rows[currRow] += c
    if (down) currRow++
    else currRow--
```

```
if (currRow == n-1) down = false
else if (currRow == 0) down = true
}
res = ""
for (0 to n) -> i {
    res += rows[i]
}
return res
```

Code:

```
if (n == 1) {
    s.o.p (s);
    return;
}
String[] rows = new String[n];
for (int i = 0; i < n; i++) {
    rows[i] = "";
}
int currRow = 0;
boolean down = true;
for (int i = 0; i < s.length(); i++) {
    char ch = s.charAt(i);
    rows[currRow] += c;
    if (down) currRow++;
    else currRow--;
    if (currRow == n-1) down = ~~true~~ False;
    else if (currRow == 0) down = true;
}
```

```
String res="";
for(int i=0; i<n; i++){
    res += rows[i];
}
S.o.p(res);
```